五字班高等代数 1 期中试题参考答案

一、对$(A\quad I)$做初等行变换有

$$\begin{pmatrix}0&1&...&1&1&0&...&0\\1&0&...&1&0&1&...&0\\...&...&...&...&...&...&...&...\\1&1&...&0&0&0&...&1\end{pmatrix}\to\begin{pmatrix}n-1&n-1&...&n-1&1&1&...&1\\1&0&...&1&0&1&...&0\\...&...&...&...&...&...&...&...\\1&1&...&0&0&0&...&1\end{pmatrix}\to$$

$$\begin{pmatrix}1&1&...&1&\frac{1}{n-1}&\frac{1}{n-1}&...&\frac{1}{n-1}\\1&0&...&1&0&1&...&0\\...&...&...&...&...&...&...&...\\1&1&...&0&0&0&...&1\end{pmatrix}\to\begin{pmatrix}1&1&...&1&\frac{1}{n-1}&\frac{1}{n-1}&...&\frac{1}{n-1}\\0&-1&...&0&-\frac{1}{n-1}&\frac{n-2}{n-1}&...&-\frac{1}{n-1}\\...&...&...&...&...&...&...&...\\0&0&...&-1&-\frac{1}{n-1}&-\frac{1}{n-1}&...&\frac{n-2}{n-1}\end{pmatrix}\to$$

$$\begin{pmatrix}1&0&...&0&\frac{2-n}{n-1}&\frac{1}{n-1}&...&\frac{1}{n-1}\\0&1&...&0&\frac{1}{n-1}&\frac{2-n}{n-1}&...&\frac{1}{n-1}\\...&...&...&...&...&...&...&...\\0&0&...&1&\frac{1}{n-1}&\frac{1}{n-1}&...&\frac{2-n}{n-1}\end{pmatrix},\ \text{因此}\ A^{-1}=\frac{1}{n-1}\begin{pmatrix}2-n&1&...&1\\1&2-n&...&1\\...&...&...&...\\1&1&...&2-n\end{pmatrix}.$$

二、若 A, B 可逆, 则$(AB)^*=|AB|(AB)^{-1}=\left(|B|B^{-1}\right)\left(|A|A^{-1}\right)=B^*A^*$. 下设 A, B 有不可逆者, 考虑方程$|\lambda I+A|=0$和$|\lambda I+B|=0$, 它们都是关于λ的不超过 n 次的方程, 其实根的个数有限, 因而存在$\delta>0$, 使得当$0<\lambda<\delta$时有$|\lambda I+A|\neq 0$和$|\lambda I+B|\neq 0$, 从而有$\left((\lambda I+A)(\lambda I+B)\right)^*=(\lambda I+B)^*(\lambda I+A)^*$, 令$\lambda\to 0$即得.

三、若不然, 则存在$x=(x_1,...,x_n)^T\neq 0$使得$Ax=0$. 设$|x_k|=\max\limits_{1\le j\le n}|x_j|\neq 0$, 由$\sum\limits_{j=1}^{n}a_{kj}x_j=0$有$|a_{kk}x_k|=\left|\sum\limits_{j\neq k}a_{kj}x_j\right|\le\left|\sum\limits_{j\neq k}a_{kj}\right||x_k|$, 即$|a_{kk}|\le\left|\sum\limits_{j\neq k}a_{kj}\right|$, 矛盾.

四、(I)有解$\Leftrightarrow r(A)=r(A,b)\Leftrightarrow r\begin{pmatrix}A^T\\b^T\end{pmatrix}=r(A^T)<r\begin{pmatrix}A^T&0\\0&1\end{pmatrix}\le r\begin{pmatrix}A^T&0\\b^T&1\end{pmatrix}\Leftrightarrow$(II)无解.

五、若不然, 则存在不全为零的$c_1,...,c_k$使得$c_1\alpha_1+...+c_k\alpha_k=0$, 不妨设$c_1\neq 0$, 由条件存在$d_1,...,d_k$使得$\beta=\sum\limits_{k=1}^{n}d_k\alpha_k=d_1\sum\limits_{k=2}^{n}\left(-\frac{c_k}{c_1}\alpha_k\right)+\sum\limits_{k=2}^{n}d_k\alpha_k=\sum\limits_{k=2}^{n}\left(d_k-\frac{c_k}{c_1}\right)\alpha_k$, 这表明$\beta$可

以只由$\alpha_2,...,\alpha_n$表出, 矛盾.

六、当$c\neq 0$时, 对 A 做初等行列变换有$\begin{pmatrix}1 & \frac{1-a}{c}\\ 0 & 1\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}1 & \frac{1-d}{c}\\ 0 & 1\end{pmatrix}=\begin{pmatrix}1 & 0\\ c & 1\end{pmatrix}$, 于是

$A=\begin{pmatrix}1 & \frac{a-1}{c}\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ c & 1\end{pmatrix}\begin{pmatrix}1 & \frac{d-1}{c}\\ 0 & 1\end{pmatrix}$; 若$c=0$, 由于$\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}a & b\\ 0 & d\end{pmatrix}=\begin{pmatrix}a & b\\ a & b+d\end{pmatrix}$, 同上

可得$A=\begin{pmatrix}1 & 0\\ -1 & 1\end{pmatrix}\begin{pmatrix}a & b\\ a & b+d\end{pmatrix}=\begin{pmatrix}1 & 0\\ -1 & 1\end{pmatrix}\begin{pmatrix}1 & \frac{a-1}{a}\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ a & 1\end{pmatrix}\begin{pmatrix}1 & \frac{b+d-1}{a}\\ 0 & 1\end{pmatrix}$.

七、做初等行列变换, 原式$=\begin{vmatrix}3 & 1 & 1 & ... & 1\\ -2 & 3 & 0 & ... & 0\\ -2 & 0 & 4 & ... & 0\\ ... & ... & ... & ... & ...\\ -2 & 0 & 0 & ... & n\end{vmatrix}=\begin{vmatrix}\left(3+\frac{2}{3}+...+\frac{2}{n}\right) & 1 & 1 & ... & 1\\ 0 & 3 & 0 & ... & 0\\ 0 & 0 & 4 & ... & 0\\ ... & ... & ... & ... & ...\\ 0 & 0 & 0 & ... & n\end{vmatrix}$

$=\left(3+\frac{2}{3}+...+\frac{2}{n}\right)\cdot 3\cdot ...\cdot n=n!\sum_{k=1}^{n}\frac{1}{k}$.

八、否. 取$A=\begin{pmatrix}J & I\\ J & 0\end{pmatrix},B=\begin{pmatrix}I & 0\\ 0 & 0\end{pmatrix}$, 其中 I 为二阶单位阵, $J=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}$有$J^2=0$, 此时

$ABAB=\begin{pmatrix}J & 0\\ J & 0\end{pmatrix}^2=0, BABA=\begin{pmatrix}J & I\\ 0 & 0\end{pmatrix}^2=\begin{pmatrix}0 & J\\ 0 & 0\end{pmatrix}\neq 0$.

注: 若 A, B 均为二阶方阵, 则答案为肯定的. 事实上, 若$B=0$结论显然; 若 B 可逆, 则有

$BABA=B(ABAB)B^{-1}=0$; 下设$r(B)=1$, 则存在可逆 P, Q 使得$B=P\tilde{B}Q,\tilde{B}=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}$,

条件化为$AP\tilde{B}QAP\tilde{B}Q=0$, 记$QAP=\tilde{A}=\begin{pmatrix}a_1 & a_2\\ a_3 & a_4\end{pmatrix}$, 则$0=\tilde{A}\tilde{B}\tilde{A}\tilde{B}=\begin{pmatrix}a_1^2 & 0\\ a_1a_3 & 0\end{pmatrix}$, 只能

$a_1=0$. 因此$\tilde{B}\tilde{A}\tilde{B}\tilde{A}=\begin{pmatrix}a_1^2 & a_1a_2\\ 0 & 0\end{pmatrix}=0$, 进而$BABA=P\tilde{B}\tilde{A}\tilde{B}\tilde{A}P^{-1}=0$.